Zeyu Yun
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Education

- **University of California, Berkeley** — Berkeley, CA
  PhD in Computer Science, advisor: Bruno Olshausen , Bin Yu — *August 2024 - Present*
- **University of California, Berkeley** — Berkeley, CA
  M.Sc. in Electrical Engineering & Computer Science; GPA: 4.00/4.00 — *August 2023 - August 2024*
- **University of California, Berkeley** — Berkeley, CA
  B.A. in Computer Science and Applied Math with Highest Honors; Major GPA: 3.93/4.00 — *Aug 2017 - Dec 2021*

Work & Research Experience

- **Apple** — Cupertino, CA
  Research Intern — *May 2026 - August 2026*
  - Trained **LLM agents** for agentic tool use with Reinforcement Learning with Verifiable Rewards **(RLVR)**, studying how post-training data scale and task difficulty affect the emergence of agentic skills.
  - Developed a synthetic data generation framework by modeling agentic tasks as graphs of references and state dependencies, automatically constructing challenging tool-use tasks for RLVR training. Improved LLM agents' task success rate on **AppWorld**, a benchmark for multi-app agentic tool use, from 73.8% to 83.3% on Normal and 59.2% to 62.6% on Challenge with RLVR and synthetic tasks. **Paper in preparation.**
- **Berkeley AI Research (BAIR) & Redwood Center, UC Berkeley** — Berkeley, CA
  Graduate Student Researcher — *May 2020 - Present*
  - Developing a **plug-and-play RLVR benchmark** that separates reasoning-mode switching from genuine skill acquisition. We find that much of the performance gain from RL comes from eliciting existing reasoning behavior, while continued training produces slower but genuine acquisition of new skills.
  - Built a unified evaluation suite for continual learning in AI agents, benchmarking prompt-based adaptation (GEPA, ACE), supervised fine-tuning (SFT), and reinforcement learning methods (GRPO, SDPO) under a common experimental framework. Under review at **ICLR 2027**.
  - Discovered that diffusion models exhibit the same visual illusion as humans, then opened the black box to identify the recurrent circuit in diffusion model that explains this behavior and how diffusion models generalize. Finally, we provided a concrete, testable hypothese for functional connectivity in the primary visual cortex during perceptual inference. Under review at **ICLR 2027**.
  - Designing a hardware-native **diffusion model** for fully analog neuromorphic chips, enabling general-purpose AI inference at ultra-low power consumption. Under review at **ICLR 2027**.
- **Flatiron Institute** — New York, NY
  Research Intern — *May 2025 - August 2025*
  - Opening the black box of diffusion models by building tools to visualize the hierarchical representations of natural images in diffusion models.
- **Redwood Center for Theoretical Neuroscience, UC Berkeley** — Berkeley, CA
  Research Assistant — *May 2020 - May 2024*
  - Building a highly compact video generative model inspired by the biological visual system that factorizing motion and form. Accepted as a poster at **COSYNE 2025**.
  - Developed **transformer-based** unsupervised learning model for generic high-dimensional data, achieving state-of-the-art performance across many diverse data modalities (AI for science). Published at **ICLR 2025.**
  - Constructed a minimalist self-supervised learning model based on neural computation principles. Our two-layer model was competitive with deep learning methods using 30+ layers. Published at **ICLR 2023 (Spotlight)**.
  - **First** to visualize large language models' (LLMs) internal representations using sparse coding, which later becomes one of the most popular tools **(SAE)** in LLM mechanistic interpretability and AI safety, followed up by major AI labs such as Anthropic and OpenAI. Published at **DeeLIO Workshop, ACL 2021**.

- **Aizip, Inc.** Cupertino, CA
  Machine Learning Engineer *Mar 2022 - Aug 2023*
  - Led development of six industrial computer-vision projects, including defect/anomaly detection, package-delivery cameras, an automated data-mining tool, a crowd-counting model, and document-signing fraud alerts.
  - Provided TinyML solutions to leading semiconductor/IC firms, including Renesas and Analog Devices.
  - Developed production-grade predictive-maintenance models deployed on large-scale assembly lines; recognized with the **Best of Sensors Award** at **Sensors Converge 2023**.

Publications & Conference

[1] ***Z. Yun**, *Y. Chen, B. Olshausen, and Y. LeCun. Transformer visualization via dictionary learning: contextualized embedding as a linear superposition of transformer factors. In *ACL workshop on Proceedings of Deep Learning Inside Out*, 2021.

[2] Y. Chen, **Z. Yun**, Y. Ma, B. Olshausen, and Y. LeCun. Minimalistic unsupervised representation learning with the sparse manifold transform. In *The Eleventh International Conference on Learning Representations (ICLR)*, 2022 (spotlight).

[3] **Z. Yun**, J. Zhang, B. Olshausen, Y. LeCun, and Y. Chen. Urlost: Unsupervised representation learning without stationarity or topology. In *The Thirteenth International Conference on Learning Representations (ICLR)*, 2025.

[4] **Z. Yun**, C. Kymn, G. Chuang, Y. Chen, and B. Olshausen. Predictive and invariant representations via motion and form factorization in natural scenes. *Computational and Systems Neuroscience (COSYNE)*, 2025.

[5] **Z. Yun**, G. Chuang, D. Dong, and Y. Chen. Denoising for manifold extrapolation. *NeurIPS Workshop on Science for Deep Learning*, 2025.

[6] C. Kymn, C. Bybee, **Z. Yun**, D. Kleyko, and B. Olshausen. Oscillator associative memories facilitate high-capacity, compositional inference. In *NeurIPS Workshop on New Frontiers in Associative Memories*.

[7] V. Lu*, Truong A.*, **Z. Yun***, and Y. Chen. Towards whitebox generative models: Scaling non-parametric sampling with representation. *arXiv preprint arXiv:2510.22196*, 2025.

[8] V. DuTell, **Z. Yun**, A. Harrington, Y. Fang, M. Hamilton, C. Koevesdi, E. Adelson, W. Freeman, and R. Rosenholtz. Gramstattexnet: Efficient, interpretable, and neuro-inspired spatiotemporal texture. In *Conference on Neural Information Processing Systems (NeurIPS)*, 2026 (spotlight).

[9] **Z. Yun**, A. Belsten, D. Bi, Z. Kadkhodaie, Y. Chen, and B. Olshausen. Toward a mechanistic understanding of inference in visual cortex and diffusion models. *The Fifteenth International Conference on Learning Representations (in submission)*, 2026.

[10] A. Harrington, N. Saxena, M. Murphy, A. Borovykh, **Z. Yun**, S. Kamath, A. Kyi, T. Darrell, J. Malik, and Y. Bai. When does continual learning require learning. *The Fifteenth International Conference on Learning Representations (in submission)*, 2026.

Honors & Awards

- Best of Sensors Award in AI/ML Category - Sensors Converge July 2023
- Honors Program - Berkeley Mathematics, Computer Sciences, UC Berkeley Aug 2019 - Dec 2021
- Upsilon Pi Epsilon (UPE) - Honor Society for Computing and Information Disciplines Aug 2019 - Dec 2021

Skills

- Programming: Python, C/C++, MATLAB
- Machine Learning: LLM post-training, Reinforcement Learning with Verifiable Rewards (RLVR), agentic tool use, synthetic data generation, continual learning, supervised fine-tuning (SFT), GRPO, SDPO, diffusion models, mechanistic interpretability, self-supervised learning
- Frameworks & Systems: PyTorch, PyTorch Lightning, Hugging Face Transformers, TensorFlow, OpenCV, Git, LaTeX, Slurm, DistributedDataParallel (DDP), multi-GPU/distributed training

Additional Activities

- Reviewer / Program Committee: NeurIPS, ICML **(Silver Reviewer Award)**, CVPR, ICLR, AAAI (Program Committee), CPAL, and ACM Computing Surveys; NeurIPS workshops on **Trust-AI-Eval**, **Mechanistic Interpretability**, and **Scientific Methods for Understanding Deep Learning**.
- Contributor for repositories Huggingface, Transformer Vis, and Word-embeddings-benchmarks.